An SIS Model for the Effect of Invasive Lionfish on Coral Reefs

Project Proposal

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1 Motivation

Supporting more species per unit area than any other marine environment, coral reefs are among the most valuable ecosystems in the world [1]. Healthy coral reefs are important for sustaining commercial and subsistence fisheries through providing thousands of species of fish for harvesting [2]. Moreover, by serving as shoreline buffers to significant amounts of energy from storms, wind, and waves, healthy corals ensure that people and habitats are protected from erosion and property damage [1].

Despite their abundant contributions to human health and happiness, coral reefs are threatened by various anthropogenic activities. Pollution, habitat destruction, and the introduction of invasive species can force reefs out of stable coral-dominated states [3]. This project will chiefly examine the effects of lionfish invading coral habitats.

As an invasive species, lionfish have few predators and can propagate rapidly in a given region [4]. As generalist feeders, lionfish primarily wreak havoc on coral ecosystems by consuming and thereby significantly reducing herbivorous populations [4]. Indeed, researchers have estimated that a single invasive lionfish can decrease recruitment of native fish by 79 percent [5]. In high productivity marine environments such as open-ocean-facing reefs, reductions in herbivorous feeders enable algal blooms, which interfere with coral recruitment, suppress coral growth and fecundity, and can directly kill certain species of coral via oxygen depletion [3]. This relationship can be seen in Figure 1. As effects traverse trophic cascades, the loss of corals is strongly associated with a decline in reef biodiversity [3]. Ensuring the survival of healthy corals is necessary for conserving the great biodiversity of our reefs.

Such damages have direct effects on communities located near coastal areas, as well as on industries that rely on tourism and fisheries. Mitigating the effects of invasive lionfish is therefore a high priority for many groups. Although efforts to reduce lionfish populations have often been spearheaded by individual spear fishers, researchers are now exploring how market-based strategies must be considered to efficiently control growing populations [6, 4].

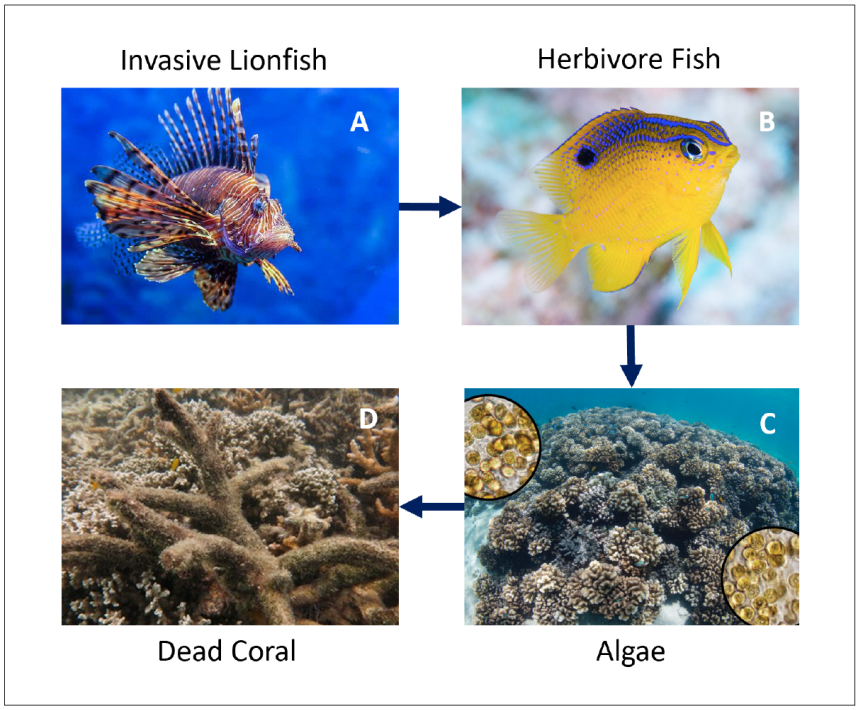

Figure 1: Relationship between the invasive lionfish population and coral reefs. Lionfish (A) are known predators of herbivore fish (B) that consume algae (C) on coral reefs. Overgrowth of algae on coral creates a hypoxic environment that eventually leads to the death of corals (D). Overpopulation of lionfish in this ecosystem leads to the loss of coral reefs.

2 Background and Question

Compartmental models are a powerful tool for analyzing epidemics. These discrete-transition models can provide useful insights about the spread of disease. For example, Susceptible-Infected-Recovered (SIR) models describe the disease driven transitions between susceptible, infected, or recovered classes for individuals in a population. These models consist of Ordinary Differential Equations (ODEs) which represent the dynamics of specified subpopulations. Specifically, an SIR model is composed of three ODEs to represent the dynamics of the susceptible class, infected class, and recovered class:

$$\frac{dS}{dt} = -\beta SI,$$
$$\frac{dI}{dt} = \beta SI - \gamma I,$$
$$\frac{dR}{dt} = \gamma I$$

Here, the entire population is the sum of individuals in the susceptible (S), infected (I), and recovered (R) classes (e.g. $S+I+R=1$). Additional parameters represent specific qualities of a given disease. The parameter β represents the rate of infection, and γ represents

the recovery rate of an individual. The SIR model can be expanded to account for vaccination strategies, birth-rate (continually introducing new individuals to the susceptible class), and death-rate (individuals leaving the population due to natural or disease driven death).

Additionally, we can consider the Susceptible-Infected-Susceptible (SIS) Model in which there are only two classes: susceptible and infected. Here, individuals transition between the two classes. Generally, an SIS model has two ODEs that are defined by the following equations:

$$\frac{dS}{dt} = -\beta SI + \gamma I$$
$$\frac{dI}{dt} = \beta SI - \gamma I$$

Here, the entire population is the sum of individuals in the susceptible (S) and infected (I) classes (e.g. $S+I=1$). Similar to the SIR model, β represents the rate of infection, and γ represents the recovery rate of an individual.

While compartmental models such as SIR and SIS are useful for modeling epidemics, they can also be applied to a broader class of problems in ecological systems, see for example [7]. This project will explore the interplay between an SIS model and a fishery harvesting model as it relates to the invasive lionfish and coral reef health.

Models indicate that commercially harvesting lionfish at rates above the maximum sustainable yield may substantially reduce population sizes [6, 8]. Researchers propose that fisheries should harvest lionfish at the Optimum Lionfish Yield (OLY), a target that exceeds Maximum Sustainable Yield (MSY) [6]. The Maximum Sustainable Yield (MSY) describes the long-run harvest that does not push the species population to extinction [8]. To ensure the survival of a species, fishers typically must harvest at rates below the MSY. However, because invasive lionfish are extraordinarily robust to typical fishing pressures, the OLY provides a rate of harvest above the MSY at which lionfish can be sustainably harvested while also reducing their impact on ecosystems [6]. The OLY-MSY comparison can be seen in Figure 2. Indeed, their model suggests that fishing at the OLY would reduce lionfish populations by an additional 42 percent relative to fishing at the MSY [6]. Despite these projected successes, fisheries presently have little incentive to fish at levels beyond the MSY, as increased fishing efforts would be required for the same yield [6]. However, through our project, we seek to demonstrate how harvesting at OLY levels may increase overall productivity through efficiently reducing invasive lionfish populations and protecting healthy coral populations.

Ultimately, using an SIS model, we aim to describe healthy / infected / dead corals in relation to the harvesting rate of lionfish. We seek to address the following questions:

1. Why should we associate the lionfish population with the rate of infection of coral reefs?

2. Might increased fishing of lionfish lead to unintentional overfishing of other fish species? What effects might this have on coral populations?

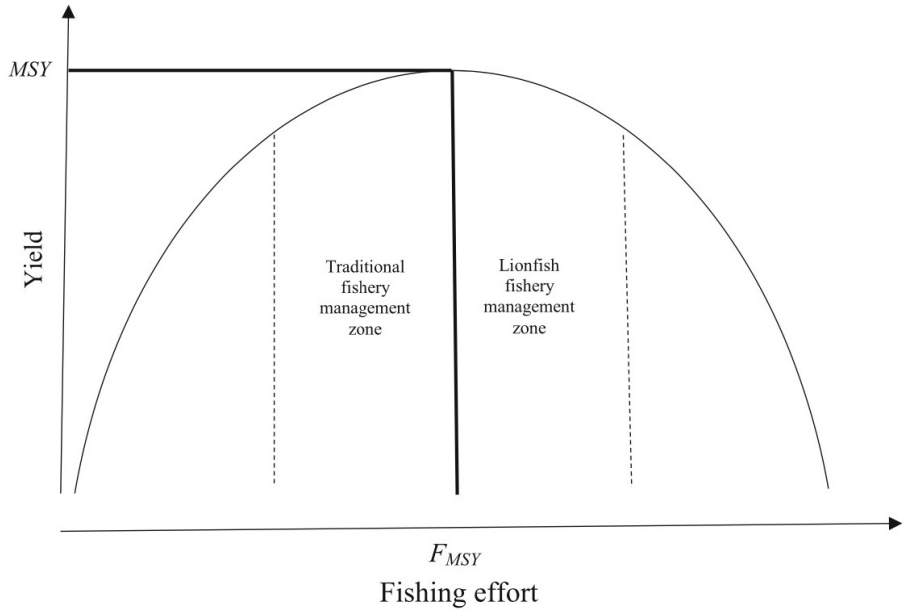

Figure 2: A generalized sustainable yield curve depicting the approximate zones of traditional fishery management targets compared to proposed lionfish fishery management targets. Figure from [6].

3. How can we find innovative ways of fishing to only significantly reduce the lionfish population?

3 Approach

Consider a population of size N, and let μ and ν be the birth and death rates, respectively, of this population. The SIS model with life dynamics (taking into consideration the birth and death rates) is given by the following equations:

$$\frac{dS}{dt} = \mu N - \beta \frac{SI}{N} + \gamma I - \nu S,$$
$$\frac{dI}{dt} = \beta \frac{SI}{N} - \gamma I - (\nu + \alpha) I.$$

Normalizing to make the total population equal to one gives the following equations:

$$\frac{dS}{dt} = \mu - \beta SI + \gamma I - \nu S,$$
$$\frac{dI}{dt} = \beta SI - \gamma I - (\nu + \alpha) I.$$

Here, S is the population of susceptible individuals, while I is the population of infected. The rate of recovery is γ, meaning that the rate of transition from the infected to the susceptible class if γI. Finally, following [9], α is the rate of death due to infection.

For this project, S will represent healthy corals, I will represent infected corals (coral suffering with hypoxia), and D will represent dead corals. Here, instead of considering a constant rate of infection β, we consider this value to be a function f dependent on the lionfish population L. This function should reflect the effects of an overpopulation of lionfish

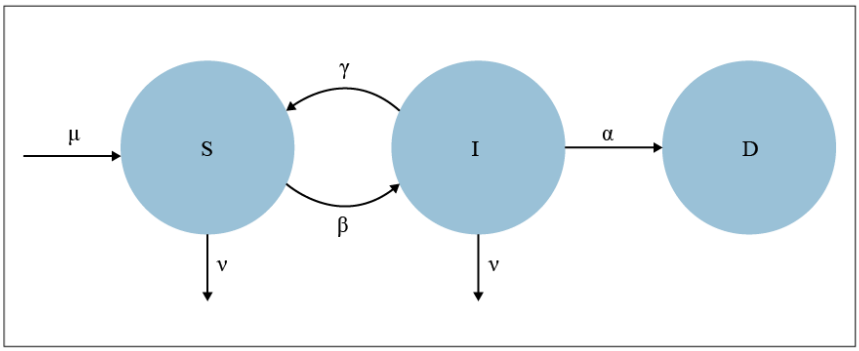

Figure 3: In this figure we see the transition between the compartments of the SIS model. For this project, the infection rate β will be a function of the lionfish population.

in the health of coral reefs, while also taking into consideration the effects of fishing over the Maximum Sustainable Yield over the dynamics of the coral population. We will investigate desirable properties of f as a function of L to adequately represent the effects of lionfish in the health of corals.

We will use a programming language such as Julia or Python to run simulations of this model. We will test different functions $f(L)$ and study the behaviour of the coral reef populations for each of these functions.

4 Member's Responsibilities

Our project will mainly focus on the development of a mathematical model that explores the relationship between invasive lionfish and coral reefs. Mirae will be responsible for investigating MSY-OLY and how this will be implemented in the model. Maddie will be responsible for investigating model parameters. Charlie will be responsible for studying the equilibria of the system and conditions on $f(L)$ for stability of the SIS model. Simeel will be responsible for developing the beta functions for the SIS model. Lastly, Marcela will be responsible for the computational simulations of the model. We will have bi-weekly group meetings to discuss our findings.

All members of the team will be responsible for finding adequate literature to better understand the effects of fishing lionfish in other populations of fish, and in general, the complexity of the ecosystem as a whole.

References

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